

Artificial Intelligence & Machine Learning – Task 3

Deliverables

This document contains the mandatory and optional deliverables completed for Task-3: Model Validation, Overfitting Control & Hyperparameter Tuning using Machine Learning.

1. Mandatory Deliverables

Jupyter Notebook implementation Cross-validation results Hyperparameter tuning results Model comparison table 2–3 page explanation report

2. Cross-Validation Results

Cross-validation was applied using `cross_val_score()` with 5-fold validation. The model achieved stable performance across multiple folds, confirming good generalization ability.

3. Hyperparameter Tuning Results

Hyperparameter tuning was performed using `GridSearchCV`. The following parameters were tested: `max_depth = [3, 5, 7, 10]` `min_samples_split = [2, 5, 10]` The tuned model showed improved performance and reduced overfitting.

4. Model Comparison Table

Model	RMSE	R2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Tuned Decision Tree	0.645430	0.682099

5. Overfitting Analysis

The Decision Tree Regressor initially showed signs of overfitting because the training performance was significantly better than the testing performance. To reduce overfitting, hyperparameter tuning techniques such as `max_depth` and `min_samples_split` were applied using `GridSearchCV`. This improved the model's ability to generalize on unseen data.

6. Tuning Approach

Hyperparameter tuning was performed using `GridSearchCV` with 5-fold cross-validation. Different combinations of `max_depth` and `min_samples_split` were tested to identify the best performing Decision Tree model. The evaluation metric used was RMSE.

7. Final Model Justification

Among all the models tested, the Tuned Decision Tree Regressor achieved the best performance with the lowest RMSE and highest R^2 score. Cross-validation results confirmed the reliability of the model. Therefore, the Tuned Decision Tree model was selected as the final model for deployment.

8. Optional Deliverables

Additional visualization plots Saved trained model using joblib